Simulation Calculation Table

Use this table to explain how the measured CSV power data is converted into energy, scaled to HPC centers, inserted into pandapower, and evaluated with grid metrics.

Step	Calculation	Formula / Logic	Meaning	How to explain it
1	Read CSV data	timestamp, gpu_power_w, cpu_power_w, gpu_utilization_%	The CSV contains measured workload values over time.	I start with measured ML workload data. Every row is one measurement point.
2	Sampling interval	Delta t = timestamp[i+1] - timestamp[i]	Finds how much time each CSV row represents.	I need the time distance between rows because energy is power multiplied by time.
3	Convert time to hours	Delta t_hours = Delta t_seconds / 3600	Energy is calculated in Wh or kWh, so seconds are converted to hours.	If sampling is every 5 seconds, each row represents 5/3600 hours.
4	Node power	P_node(t) = P_GPU(t) + P_CPU(t)	Combines GPU and CPU power into one node demand.	One compute node consumes GPU power plus CPU/system power.
5	Energy per CSV row	E_row_Wh = P_node_W(t) * Delta t_hours	Calculates the energy represented by one row.	Each row gives power for a small time interval, so I multiply power by that interval.
6	Total energy	E_total_Wh = sum(E_row_Wh)	Adds all row energies to get workload energy.	The total energy is the sum of all small energy pieces in the CSV.
7	GPU utilization	u(t) = GPU_utilization(t) / 100	Converts utilization from percent to 0-1.	Utilization tells how busy the GPU is at each timestamp.
8	GPU power limit	P_GPU,limited(t) = P_GPU(t) * [1 - u(t) * r]	Power limit effect depends on utilization.	High GPU utilization gives stronger power reduction; low utilization gives smaller reduction.
9	Limited node power	P_node,limited(t) = P_GPU,limited(t) + P_CPU(t)	Rebuilds node power after scenario change.	After changing GPU power, I add CPU power again to get new node demand.
10	Scale to one HPC center	P_center_IT(t) = P_node(t) * N_nodes	Scales one measured node to many nodes.	One measured node is multiplied by the number of nodes in the center.
11	Apply PUE	P_facility(t) = P_center_IT(t) * PUE	Adds cooling and facility overhead.	The grid supplies the whole data center, not only GPUs and CPUs.
12	Scale to many centers	P_total(t) = P_facility(t) * N_centers	Simulates several HPC centers.	For 20 centers, the facility demand is multiplied by 20.
13	Convert W to MW	P_MW(t) = P_total_W(t) / 1,000,000	Pandapower loads use MW.	Before inserting into pandapower, I convert watts to megawatts.
14	Reactive power	Q(t) = P(t) * tan(arccos(pf))	Estimates reactive power from power factor.	AC power flow needs active and reactive power, so Q is estimated from pf.
15	Update grid load	net.load.at[id, "p_mw"] = P_MW(t)	Sets the HPC demand for this timestep.	For each timestamp, the current HPC demand becomes the pandapower load.
16	Run power flow	pp.runpp(net)	Solves the AC grid equations.	Pandapower recalculates voltages, currents, transformer loading, and line loading.
17	Minimum voltage	min(net.res_bus["vm_pu"])	Lowest bus voltage in the grid.	This shows whether the HPC load causes voltage drops.
18	Transformer loading	max(net.res_trafo["loading_percent"])	Highest transformer utilization.	This shows whether a transformer becomes overloaded.
19	Line loading	max(net.res_line["loading_percent"])	Highest line/cable utilization.	This shows whether a line becomes overloaded.
20	Convergence	net.converged	Checks if the power flow solved successfully.	If it fails, the load or grid setup may be too stressed or numerically unstable.
21	Average and peak power	P_avg = mean(P_total(t)); P_peak = max(P_total(t))	Shows typical and maximum demand.	Average power describes normal demand; peak power is important for grid stress.
22	Cost	Cost(t) = P_kW(t) * Delta t_hours * price(t)	Calculates electricity cost per row.	Cost depends on energy in each timestep and the electricity price at that time.

Short summary: CSV power values -> power times time = energy -> scale to nodes/centers -> apply PUE -> convert to MW -> run pandapower for each timestamp -> evaluate voltage, transformer loading, line loading, convergence, energy, peak power, and cost.